

Plotting Geographical Data Transcript

Now we will move to the second part of today's lesson where we'll be trying to understand how to plot geographical data. As I explained in the visualisation lectures, plotting geographical data is of very high importance on certain use cases especially when you have distributed geographical presence as a business entity and these days presenting on maps brings in very much needed attention to the audience because they can closely associate that with the space in which they are living. So let us understand how to plot the geographical data or how to connect with the geographical data and bring in our visualisations onto the maps and bring more context to the phenomenon that we are trying to explain.

So the library that enables you to do this in Python environment is called Geopandas. Geopandas is nothing but geographical panda which enables you to use the geometry. There will be a point like we might get into the specifics of this there is a shapefile which has the attributes that are required to draw a map, present data on the map and we can use these features to improvise our visualisations even on the graphs. It is not just the graph but the way the graph is coloured, what is the intensity of the colour, what particular regions you want to highlight.

All these are dependent on the shape and geometry features which we will be using for plotting. So there are two ways Geopandas uses a geodata frame which is like regular pandas data frame but the geometry column is what brings in the richness to the map visualisations. So whether they are located as points, lines or polygons and then you can plot a geodata frame directly with dot plot with all the math plotlib handles that we just have seen across all our lessons.

So let us try to do this particular plotting. First for that you have to instal the geodatasets into Geopandas. So using this pip instal command so you can do that part then you would have installed the necessary files that you need.

So then what we are going to do right so we are going to yeah I am going to run a simple code here I am importing the geodatasets and geopandas okay so I'm importing these two libraries right I will first run the selection so run the selection I have imported these two and then I am naming two data frames Chicago is equal to geopandas dot read file which is the read CSV kind of a command which reads the geo file right and geodatasets it is getting the Chicago data and the groceries data from the same file and I am creating the two data frames from this okay I am running the two data frames from this so all the data has been extracted now let us see how the data is looking for Chicago so this is how the data you have community population in 2010 population in 2000 okay so this is what we have and this is the geometry data which I was mentioning it is a multi polygon with this particular attributes latitude and longitude helping it to construct the map right similarly you can also see the can also see the groceries groceries dot head so you got the y-coordinate x-coordinate so this is one more way

to showcase this x-coordinate and y-coordinate status address right which is the chain that is holding this okay now what I am doing I am simply using a basic plot Chicago dot plot it is showing the plot of Chicago state with x and y coordinates that is showing the map that is a simple representation but yeah so let us see so it is just showing geoplot a geodata frame so what are the parameters what all you can do right so geopandas so these are the features that you can attributes that you can use for your plot map plot okay what does it do generate a plot of geodata frame with my plot lip okay if a column is specified or the column will be based on values in that particular column so what I am trying to do

And now I am trying to parse a particular column into that particular plot function so I'm saying column is equal to population in 2010 so I am doing population in 2010 I just got the population in 2010 and varying upon the intensity of the population the colours are given okay so now I want to change the colour right and just doing this I am also including the legend by including legend is equal to true function and I got the legend is equal to true with darker shades indicating the darker colour okay yeah so now what I am trying to do I want to indicate the legend okay in a horizontal direction and also add a title to the legend here the legend is there which is intuitive saying that this is the intensity but I don't which parameter it is representing and it is consuming more space so I want to just do orientation as horizontal

And I want to add the labels to that so I'm just doing that so yeah so this is what we are getting here right so you like this you can explore the geographical data which is there in the these days you have lot of data sets which are geographical data and you can use the geopandas function to plot this again we will take up one more example in trying to yeah plot the geodata so let us see this one more example this is a very simple example it is already pre established data set that we are trying to do so all that you need is you need especially the geometry variable or the X or Y coordinate for plotting the maps this is very this is mandatory in some sense to plot the data in a geographical map and once you have this you can use the geopandas library for plotting the map data okay and you can use try to use the different colours and saturations to showcase that so again yeah one thing I wanted to show us a geopandas Python so so this is one thing I would sincerely urge all of you to please go through and then these are examples gallery so you have so many features right so again the universe is very large we can't deal with all this maybe we will take a couple of examples in the live session but what I would sincerely urge you is to take each example right and see how it is using the different features within the geopandas library to plot the data okay so depending upon the use case that you have for example this was the same Chicago data set that we try to do so yeah yeah so now let me ask you a question here you need not present the entire map

this is the just dot plot people who are from Chicago would understand that this is represents the state of Chicago because following the just lot principle of closure just that you have the dots which form the geographical ship that you are generally aware of you will be able to

know this is Chicago or if you plot the states in India you will know these are states in India right so like that you can use the different features in geopandas and try to get around I mean try to develop the map based data visualisations right so there are so many examples here please explore them maybe couple of them we might try to do in the live lecture okay now let us try to yeah do this particular code a matplotlib in line because what does this code it will help us to showcase the exceptions when the code will run without opening in the new cell so I have done this part so what am I doing here I have done the import of geopandas library and are not installed we have to instal them using pip instal so this is what we had to do so maybe I will do this again here or we already have the pip instal done so we have already done the pip instal here

so we are trying to run the code here so yeah so let us see this part what we are trying to do we are trying to pass data from the URL where we are getting data from this particular website okay so this is the github website so let me follow the link and just following so this is the data that we have comma separated values in particular format with the latitude longitude data so I have parsed this data like in the earlier case we have taken the data we have defined the URL and then we are now using the `geopd.read` is `geopandas.read` file from the URL okay we are reading this particular data from the `geopandas` right using the `geopandas.read` underscore file now plotting the figure okay we are plotting figure `dotting` plot dots of plots world dot plot is the world map that comes from the `geopandas` library and you are using the data that is from the file okay so you are plotting from the world dot plot we are using that particular plot from that particular data frame and sorry we are plotting the data for this data frame named as world using the world dot function and you are running the code okay so now you got that data so this is what is plotted this is the plain data the entire world is plotted it is simply grey in colour and then you have plotted all these countries that are there in the world okay now what you want to do you want to overlay a particular city okay you want to overlay particular city in this case New Delhi New York and Tokyo with this particular latitude and longitude data so what you are trying to do first you are defining a particular data frame with column city latitude

And longitude New Delhi corresponds to these two values New York to these three and Tokyo to these two sorry so then we are trying to define this particular data frame then I am saying `GDF` underscore cities is a `geopanda` data data frame where data is coming from the city's data and geometry is coming from the longitude and latitude see I am creating a geo data frame by using the latitude longitude and creating the and attaching them the city names okay so this this command `GPD` dot geo data frame creates a `geopanda` data frame and as I said earlier you need the geometry for mapping them onto a for a plotting them on a map and how is geometry coming geometry is coming from latitude and longitude and which values are present here so using this particular data frame cities underscore data or what we call it as most appropriately dictionary in Python parlance you are creating a set of arrays which are a dictionary you are parsing the dictionary to create a data frame and then creating



geometry objects using the latitude and longitude values okay so on the selection I have now created the GDF dot cities right so then finally I am plotting the GDF dot cities onto the world map and trying to show that just running this all together once again so I have plotted this labelled major cities India and New Delhi Tokyo and New York onto the world map so again the main catch in this is you have used a pre-established data set on the different continents and their different countries and their latitude and longitude onto the world map they have shown the world map using this world dot plot and then you wanted to highlight a few cities right and then you got these values of city dot New York so what I can do just to showcase in this let me put London just putting it to be latitude of London would be somewhere around if it is it is about New York so London is somewhere about New York let me put it as and somewhere yeah it let it let me try a value I don't know the exact value and this trying it to be 45.7 and I know the Greenwich Language passes through London UK putting it as 0 and then this time trying to run the code again so see okay I ended up being in Paris but still close enough this is how you create more data and try to overlay that data into the map so I might as well we name it as Paris okay yeah so this is what I have done so it's a simple thing we have created a geodata frame but the most important part here is to get the latitude and longitude and take that into the geometry of the geodata frame using which the map can be formed okay the visualisation can be showcased as a map for which you need the geometry data right we will see more examples in the live class and also we will give you a exercise for practise thank you.

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